2D Matrices: Shears

Questions 1–13

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}$

Question 2. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$. Write down the area of the shape's image.

Question 3. The matrix $\mathbf{A}$ represents a shear with the x-axis fixed, which maps $(0, 1)$ to $(k, 1)$ and $(-2, 4)$ to $(18, 4)$. Determine the value of k.

Question 4. The matrix $\mathbf{A}$ represents a shear with the y-axis fixed, which maps $(1, 0)$ to $(1, k)$ and $(22, -6)$ to $(22, -94)$. Determine the value of k.

Question 5. The triangle X has vertices at $(-1, \sqrt{2})$, $(-3\sqrt{2}, 0)$ and $(4\sqrt{2}, 0)$. X is transformed by a shear, with the x-axis fixed, with $(0, 1)$ mapped to $(8\sqrt{2}, 1)$. Using matrices, find the exact coordinates of the vertices of triangle X' and hence show that the area of X is the same as the area of X'.

Question 6. The matrix $\mathbf{A}$ represents a shear, with the y-axis fixed, with $(8, 3\sqrt{3})$ mapped to $(8, 5\sqrt{3})$. Find the matrix $\mathbf{A}$.

Question 7. The matrix **A** represents a shear, with the x-axis fixed, with $(50, 4)$ mapped to $(4, 4)$. Find the matrix **A**.

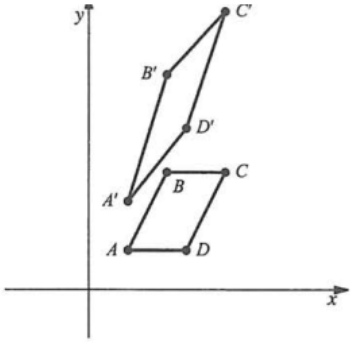
Question 8. The matrix **A** represents a shear, with the x-axis fixed, with $(45, 9)$ mapped to a point that lies on the y-axis. Find the matrix **A**.

Question 9. The matrix $\mathbf{A}$ represents a shear, which maps the point $(5, 0)$ to itself, and the point $(3, 20)$ to $(11, 20)$. Find the matrix $\mathbf{A}$.

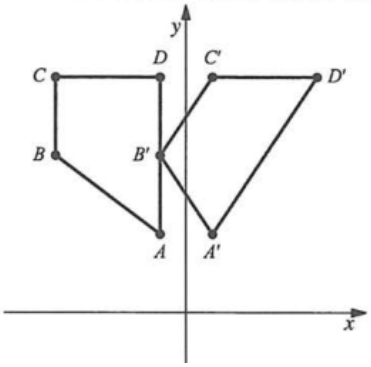
Question 10. The matrix $\mathbf{A}$ represents a shear, which maps the point $(-12, 8)$ to $(-12, 17)$. The matrix $\mathbf{B}$ represents a shear, which maps the point $(-12, 17)$ to $(-12, 8)$.

a) Find the matrix $\mathbf{A}$.
b) Find the matrix $\mathbf{B}$.
c) Show that $\mathbf{AB} = \mathbf{I}$.

Question 11. The vertices of a parallelogram $ABCD$ have coordinates $A(2,2)$, $B(4,6)$, $C(7,6)$ and $D(5,2)$. The parallelogram is mapped to quadrilateral $A'B'C'D'$, as shown in the diagram below, by the shear associated with matrix $\mathbf{M}$. The coordinates of D' are $(5, 8.25)$. Find the matrix $\mathbf{M}$ and the coordinates of A', B' and C'.



Question 12. The vertices of a trapezium $ABCD$ have coordinates $A(-1,3)$, $B(-5,6)$, $C(-5,9)$ and $D(-1,9)$. The parallelogram is mapped to the trapezium $A'B'C'D'$, as shown in the diagram below, by the shear associated with the matrix $\mathbf{M}$. The coordinates of D' are $(4,5)$. Find the matrix $\mathbf{M}$ and the coordinates of A', B' and C'.



Question 13. Given $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$, where k is a constant,

a) describe the linear transformation associated with the matrix $\mathbf{A}$
b) Find $\det \mathbf{A}$
c) By considering $\det \mathbf{A}$, describe two properties of the transformation associated with the matrix $\mathbf{A}$.